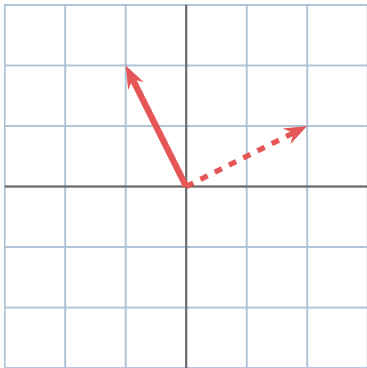


Rotation by 90°

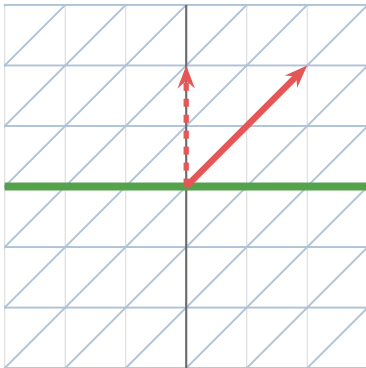
$$\lambda^2 + 1 = 0$$



no real eigenvalues:
every vector is turned off its line

Shear

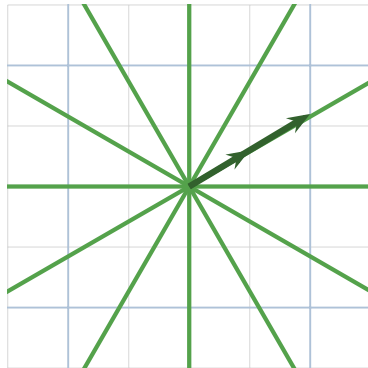
$$(1 - \lambda)^2 = 0$$



$\lambda = 1$ only: the x-axis is
the one line of eigenvectors

Scaling by 2

$$(2 - \lambda)^2 = 0$$



$\lambda = 2$ only, but every vector
is an eigenvector

grey: grid before blue: grid after dashed arrow: a vector before solid arrow: after green: lines of eigenvectors